

Article

One Model, Many Arms: Amortized Multimodal Inverse Kinematics Across a Continuous Family of Soft Continuum Manipulators

Adhvay Jagadeesh ^{1,*}

¹ Independent Researcher

* Correspondence: [\[email\]](#)

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Abstract: A learned inverse-kinematics (IK) model for a soft continuum arm is trained for one arm. Yet no two fabricated soft arms are the same arm: segment lengths, stiffness, and actuation gain vary by fabrication tolerance, and a model fitted to one instance is silently wrong on the next. We ask whether a single conditional diffusion model can amortize inverse kinematics across a continuous family of soft arms—solving IK for morphologies it never saw—and what makes that possible. On a piecewise-constant-curvature pneumatic arm whose four segment lengths and four curvature gains are each drawn per sample from a continuous box, a diffusion model conditioned on the morphology vector reaches 0.39 mm best-of-32 tip error with 100% success on held-out in-distribution morphologies—matching a specialist trained on 5×10^5 samples of its own arm, with zero samples of any specific arm. A controlled ablation isolates the cause: the identical randomized data with the morphology *withheld* yields 12.9 mm and 12% success, a $33\times$ gap. Domain randomization fails not because randomizing is wrong but because hiding the morphology forces the model to regress to the mean arm, which also explains why gain randomization recovers nothing under dynamics-model change. Generalization is flat across the training box and degrades gracefully for 10–25% beyond it before a sharp cliff. Deterministic and mixture-density regressors trained on the same data do not amortize (14–16 mm in-distribution), because the morphology-conditioned inverse map is more multimodal, not less. We report the single-arm advantage, its redundancy mechanism, and the limits of transfer under dynamics-model change with three-seed statistics throughout. Code, data, and per-experiment result files are released.

Keywords: soft robotics; continuum manipulators; inverse kinematics; diffusion models; amortized inference; domain randomization; multimodal learning

1. Introduction

A multi-segment pneumatic continuum arm with three chambers per segment has more actuation degrees of freedom than task degrees of freedom: for a four-segment arm, a 12-D actuation vector determines a 3-D tip position. The inverse map is one-to-many, and its preimages are not a connected null space—distinct arm shapes reach the same tip. A deterministic learned IK regressor trained by motor babbling [1] fits the conditional *mean* of this multimodal distribution and can return an actuation that realizes none of the true solutions. A conditional diffusion model [2,3] learns the full conditional distribution $p(\mathbf{q} | \text{target})$ and can sample every solution mode. We establish below, with three-seed statistics, that this advantage is real, that it is specifically a redundancy effect, and that it does not survive a change of dynamics model.

Every learned soft-arm IK method we are aware of, including the single-arm models we begin with, trains one model for one arm. This paper starts from the observation that *one arm* is a fiction. A

soft manipulator is a fabricated object: its segment lengths depend on where discs were bonded, its bending stiffness on the backbone rod’s diameter (which goes as the fourth power, so a $\pm 4\%$ diameter tolerance is a $\pm 17\%$ stiffness range), and its actuation gain on tendon routing and material batch. Two arms built from the same drawing are two different arms. A model that must be retrained for each one—at 5×10^5 simulated samples and half an hour per arm—is a model that does not scale past the bench it was trained on.

The question is therefore whether a single model can *amortize* inverse kinematics across a family of arms: trained once on a continuous distribution of morphologies, and asked to solve IK for a specific morphology it never saw, given only that morphology’s parameters as a conditioning input. This is the soft-robotic instance of a question now central to rigid-robot learning, where diffusion IK has been made to generalize across kinematic-tree *topologies* [4,5]. Soft arms pose it differently: the morphology is not a discrete topology but a continuous vector, and the conditional distribution the model must represent is more multimodal, not less—the same tip is reached by different shapes on different arms.

There is a specific reason to doubt the answer is yes. Domain randomization (DR) [6]—training on randomized arm parameters so that the model is robust to their variation—recovers *nothing* under model change in our experiments, while degrading in-domain accuracy fivefold (Sec. 7). If randomizing the morphology hurts, why would a randomized family be learnable at all? The answer this paper gives is that DR randomized the arm but *hid* it from the model. With the morphology withheld, the model must average over an ambiguous family; with it exposed, the model can learn the inverse map as a function of morphology. The difference is one conditioning vector, and it is the difference between a 12.9 mm model and a 0.39 mm one.

Our contributions:

1. **Amortization is free.** A single morphology-conditioned diffusion model reaches 0.39 mm best-of-32 tip error and 100% success on held-out in-distribution morphologies, matching a per-arm specialist that spent the same data budget on one arm alone (Sec. 5).
2. **The information, not the data.** The same randomized training set with the morphology withheld—domain randomization—reaches 12.9 mm and 12% success. Withholding the morphology costs $33\times$ in accuracy and lands *below* the single-arm model even on that arm. This is the mechanism behind DR’s failure and the first controlled isolation of it (Sec. 5.2).
3. **A measured generalization envelope.** Accuracy is flat across the training box, degrades gracefully 10–25% past its edge, then cliffs; extrapolation is asymmetric—shorter and stiffer transfer better than longer and more compliant (Sec. 5.3).
4. **Only a multimodal model amortizes.** Deterministic and mixture-density regressors trained on the identical morphology-conditioned data reach 14–16 mm in-distribution. Amortization raises the multimodality of the inverse map, and only a model that represents it can exploit the conditioning (Sec. 5.6).
5. **The single-arm advantage and its limits, with three-seed statistics.** Diffusion IK dominates MLP and MDN baselines on one arm, recovering 0.84 ± 0.03 of ground-truth modes where both recover none; the advantage is isolated as redundancy resolution by two controlled manipulations; and the learned inverse map collapses under dynamics-model change on every seed, with the recovery mechanisms quantified (Secs. 4, 6, 7). The amortization result is read against this honest account of what a learned inverse map can and cannot do.

Negative and bounded results are reported with the same prominence as positive ones; Sec. 9 states the limitations explicitly.

2. Related Work

IK for continuum arms. Non-learned approaches solve continuum IK by optimization through a model at query time: Jacobian schemes for nonconstant-curvature arms [1], real-time FEM inversion [7], and metaheuristic search with obstacle constraints [8]. The constant-curvature kinematics underlying our simulator are reviewed in [9]; STIFF-FLOP-style pneumatic arms motivate the three-chamber

actuation model [10]. Learned approaches fit a deterministic target-to-actuation map on motor-babbling data [1]; mixture density networks [11] are the classical answer to multimodal regression and serve as our learned multimodal baseline. Every learned method in this line is trained for one arm.

Diffusion and flow models for robots. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [12] underpin diffusion policies [13]. IKDiffuser [4] solves IK for rigid kinematic trees and generalizes across *topologies* by tokenizing end-effectors; generative graphical IK [5] likewise generalizes over rigid-robot graphs. Neither addresses compliant bodies, and both treat morphology as a discrete structure rather than a continuous parameter. For continuum robots, flow matching has recently been used to learn a *forward* shape model conditioned on actuation across multi-module tendon-driven arms [14]; our problem is the inverse, and the quantity we condition on is the morphology, not the actuation.

Domain randomization and amortized inference. DR [6] trains on randomized simulator parameters to induce robustness; the parameters are, by design, not exposed to the policy. The alternative—exposing the parameters as conditioning so that a single model serves a family—is amortized inference in the sense of [15], and is the design principle behind morphology-conditioned locomotion policies [16]. We are not aware of it having been applied to soft-arm IK, nor of a controlled comparison that isolates exposing the parameters from randomizing them.

3. Setup

3.1. The arm family

The training simulator is a four-segment piecewise-constant-curvature (PCC) arm [9]. Segment i has length L_i and three 120° -spaced chambers with normalized pressures $\mathbf{q} \in [0, 1]^{12}$; a linear chamber-to-curvature map with gain g_i yields per-segment curvature components (k_x, k_y) , and forward kinematics composes constant-curvature arcs. The model is batched and differentiable in $\mathbf{q}$, and—new here—in the per-segment lengths and gains, so that one forward pass evaluates a batch of *different* arms.

A *morphology* is the vector $\boldsymbol{\theta} = (L_1, \dots, L_4, g_1, \dots, g_4) \in \mathbb{R}^8$. The *nominal* arm of Sec. 4 is $L_i = 60$ mm, $g_i = 25$ (total length 240 mm). The *training box* is $L_i \in [40, 80]$ mm, $g_i \in [15, 35]$, each drawn independently and uniformly per sample; the nominal arm lies inside it. Total arm length across the box spans 160–320 mm and maximum reach 120–353 mm among the test morphologies below.

3.2. Three models, one training set

The design turns on a single contrast, so it is stated precisely. One training set of 5×10^5 samples is generated: for each, a morphology $\boldsymbol{\theta}$ is drawn from the box, an actuation $\mathbf{q}$ is drawn uniformly, and the tip is computed on *that* arm. Three models are then trained.

Amortized conditioned on $[\text{tip}, \boldsymbol{\theta}]$ (11-D). The model is told which arm each sample came from.

Blind conditioned on $[\text{tip}]$ only, on the *identical* samples. The model sees the same diversity of arms and is not told which one any sample is. This is domain randomization, extended from the gain-only randomization of Sec. 7 to length and gain.

Nominal conditioned on $[\text{tip}]$, trained on 5×10^5 samples of the nominal arm alone. This is the single-arm model of Sec. 4, and represents current practice.

Amortized and Blind differ in nothing but the eight conditioning dimensions. Any gap between them is attributable to the information, not to the data.

3.3. Models and protocol

The diffusion model is a residual MLP denoiser (six FiLM-conditioned blocks, width 512) trained as a DDPM ($T=1000$, cosine schedule) with classifier-free guidance (condition dropout 0.1); sampling uses DDIM with 20 steps and guidance weight 1.5; both were fixed by ablation on the nominal arm

(guidance has an inverted-U optimum at 1.5; accuracy and recall saturate by ~ 20 steps). Baselines are an MLP regressor and an MDN (eight components) matched in width, each trained in all three regimes, plus a classical optimization-based reference (gradient-descent IK through the differentiable PCC model from 32 random restarts, Adam, 300 iterations; grad-IK). Unless stated otherwise we sample $K=32$ candidates per target and report best-of- K ; success means tip error under 5 mm.

Held-out morphologies. Test arms fall in three bands. *Nominal* is the single-arm model’s arm, inside the box, so an interpolation test. *In-distribution* is five morphologies drawn from the box with a separate seed; with eight continuous parameters, no test morphology coincides with any training sample. *Extrapolation* is five arms outside the box: all segments long ($L=90$ mm), short (30 mm), stiff ($g=40$), compliant ($g=10$), and a mixed case alternating long-stiff and short-compliant segments. Every metric is computed on the test arm’s own kinematics: a PCC arm is instantiated at that morphology and the full metric suite runs on it. Accuracy uses 512 targets per morphology sampled from that arm’s reachable set.

Ground-truth modes. For mode recall we enumerate the true IK solution set by a 2×10^6 -sample sweep on the test arm, keeping actuations within tolerance and clustering with DBSCAN [17] ($\epsilon=12$, $\text{min_samples}=5$; match radius 15) in the 8-D per-segment curvature space, which removes the chamber map’s per-segment null space. Evaluation targets on the nominal arm average 2.05 modes.

Statistics. Every $\pm$ is the population standard deviation over three training seeds. Where a result is single-seed it is marked so.

4. The Single-Arm Advantage

We first establish the single-arm result the amortization study builds on: on the nominal arm, with three training seeds, does a diffusion model beat deterministic and mixture-density regressors, and is the margin a multimodality margin?

Table 1. Single-arm comparison on the nominal arm (mean $\pm$ std, 3 seeds; $K=32$). Mode recall and diversity over successful samples only. grad-IK is one optimization run per target and needs the differentiable simulator at query time. Source: `results_seed{0,1,2}.json`.

	Diffusion	grad-IK	MLP	MDN
Tip err. best-of- K (mm)	0.44 ± 0.03	0.013	13.4 ± 0.0	17.3 ± 0.2
Tip err. single sample (mm)	1.69 ± 0.15	—	13.4 ± 0.0	83.6 ± 2.7
Success @ 5 mm	1.00	1.00	0.060	0.027
Mode recall	0.84 ± 0.03	0.85	0.00	0.00
Diversity (curv. space)	28.2 ± 0.2	29.0	0.0	0.0
Obstacle success	0.978 ± 0.002	0.980	0.040	0.018
Query time (ms/target)	38.6 ± 4.9	428	0.013	0.022

Best-of- K diffusion reaches 0.44 mm at 100% success; the MLP, structurally limited to the conditional mean, plateaus at 13.4 mm, and the MDN does not improve on it. Against enumerated solution sets, diffusion recovers 0.84 ± 0.03 of modes and both learned baselines recover none—the MDN not through component collapse but because no component is accurate enough to count as a success. When the task requires any of K samples to be simultaneously accurate and collision-free among three random spherical obstacles, diffusion succeeds on 97.8% of trials against 4.0%. grad-IK, with the simulator in hand, is the accuracy ceiling at 0.013 mm and recovers modes at a rate on par with diffusion; the learned model’s distinction is that it answers from samples alone at a $5\text{--}11\times$ lower query cost. On every task metric the *worst* diffusion seed outperforms the *best* learned-baseline seed—for tip error, 0.49 vs. 13.33 mm, a gap exceeding $400\times$ the across-seed standard deviation. The MDN’s zero recall is not a training pathology: NLL decreases monotonically, 7.8 of 8 components remain effective, and sweeping the component count over $\{4, 8, 16, 32\}$ never surpasses the plain MLP. These

conclusions are not artifacts of the enumeration hyperparameters: across a 3×3 grid of DBSCAN ϵ and match radius, the learned baselines stay at zero everywhere while diffusion and grad-IK trade the lead.

5. Amortization Across Morphologies

Table 2. The central comparison: best-of-32 tip error and success at 5 mm, mean over the morphologies in each band. Amortized and Blind were trained on the identical 5×10^5 samples; Nominal on 5×10^5 samples of the nominal arm. Seed 0; three-seed $\pm$ pending [3-seed]. Source: results_morph_3way_seed0.json.

Band	Amortized		Blind		Nominal	
	err (mm)	succ.	err (mm)	succ.	err (mm)	succ.
Nominal arm	0.43	100%	5.54	47%	0.40	100%
In-distribution	0.39	100%	12.86	12%	32.6	4%
Extrapolation	6.09	62%	34.4	2%	61.6	0.5%

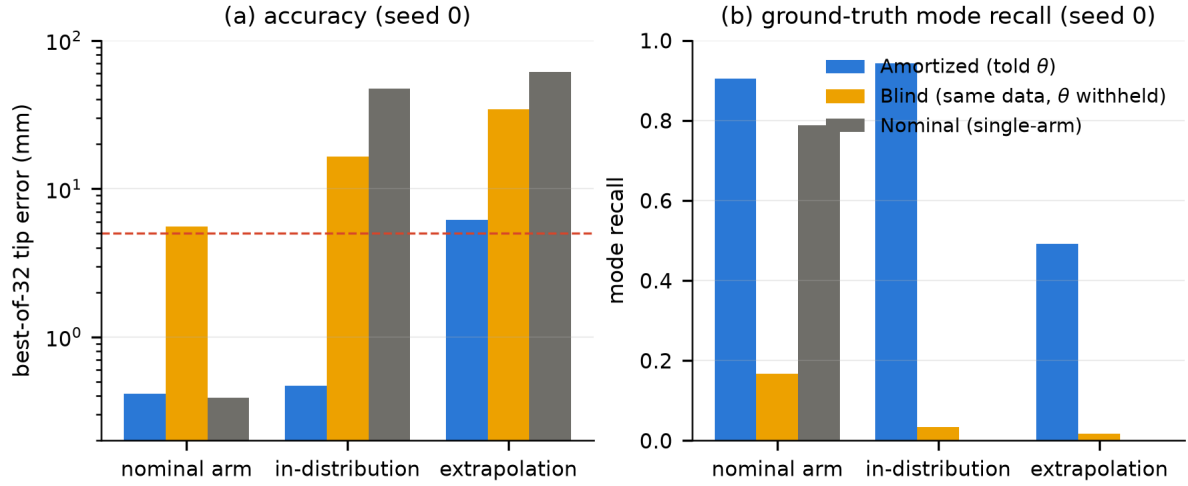


Figure 1. The central comparison. (a) Best-of-32 tip error (log scale; dashed line is the 5 mm success tolerance) and (b) ground-truth mode recall, by test band, for the three models of Sec. 3.2. Amortized and Blind share identical training data. Seed 0; three-seed error bars pending [3-seed].

5.1. Amortization is free

Table 2 and Figure 1 give the paper’s central result. On the five held-out in-distribution morphologies, the amortized model reaches 0.39 mm best-of-32 tip error with 100% success on every one (per-arm range 0.26–0.53 mm). On the nominal arm—which it never saw as an exact morphology—it reaches 0.43 mm, against 0.40 mm for the specialist that spent its entire 5×10^5 -sample budget on that arm alone. The cost of generalizing across the family, measured on the family’s own nominal member, is 0.03 mm.

The specialist, conversely, is useless off its arm: 32.6 mm and 4% success in-distribution, 61.6 mm and 0.5% extrapolating. Its errors on individual in-distribution arms range from 10 to 65 mm; on the long-segment arm, 115 mm. A model trained on one arm has learned a lookup table for that arm, and the table is wrong everywhere else.

5.2. The information, not the data

The Blind model was trained on the identical samples as the Amortized model. It reaches 12.86 mm and 12% success in-distribution—a $33 \times$ gap in error, attributable to nothing but the eight conditioning dimensions. On the nominal arm it reaches 5.54 mm and 47%: training on a randomized family *without* the morphology lands worse than the single-arm specialist on that very arm.

This is the mechanism behind domain randomization’s failure. With the morphology hidden, a given tip target is consistent with a different actuation on every arm in the family; the model cannot resolve which, and regresses toward an actuation for the mean arm. The envelope sweep of Sec. 5.3 shows exactly that signature: the Blind model is most accurate near the centre of the box and degrades toward every edge—not because the edge arms are harder, but because they are furthest from the mean it has learned. The finding that gain-DR degrades in-domain accuracy from 0.44 to 2.18 mm and recovers nothing under model change (Sec. 7) is the same effect, observed without the ablation that explains it. DR does not fail because randomization is the wrong idea; it fails because withholding the parameters is.

5.3. The generalization envelope

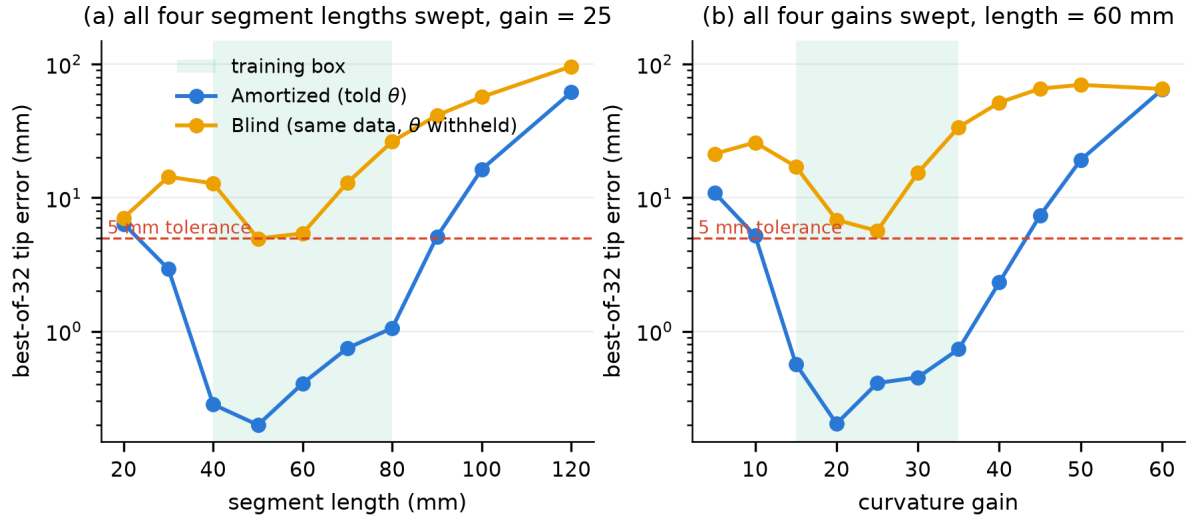


Figure 2. Generalization envelope. One morphology axis is swept with the other held at nominal, all four segments together; shaded region is the training box. The amortized model is flat inside the box, degrades gracefully for 10–25% past its edge, then cliffs. The Blind model is best at the box centre and worse toward every edge—the signature of a model that learned the mean arm. Seed 0. Source: results_envelope_seed0.json.

Table 3. Generalization envelope: one morphology axis swept with the other at nominal, all four segments together. Best-of-32 error and success. Shaded rows are inside the training box. Seed 0; source: results_envelope_seed0.json.

L	Segment length (mm), $g=25$		g	Curvature gain, $L=60$	
	Amortized	Blind		Amortized	Blind
20	6.4 / 31%	7.1 / 44%	5	10.9 / 20%	21.3 / 13%
30	2.9 / 94%	14.4 / 1%	10	5.2 / 56%	26.0 / 12%
40	0.28 / 100%	12.8 / 7%	15	0.57 / 100%	17.1 / 14%
50	0.20 / 100%	5.0 / 57%	20	0.20 / 100%	6.8 / 36%
60	0.41 / 100%	5.4 / 50%	25	0.41 / 100%	5.6 / 41%
70	0.75 / 100%	13.0 / 12%	30	0.45 / 100%	15.4 / 6%
80	1.06 / 99.6%	26.4 / 1%	35	0.73 / 99.6%	33.8 / 1%
90	5.1 / 60%	41.4 / 0.4%	40	2.3 / 98%	51.7 / 0%
100	16.3 / 3%	56.8 / 0%	45	7.4 / 30%	65.6 / 0%
120	61.6 / 0%	96.2 / 0%	50	19.2 / 5%	70.1 / 0%
			60	64.5 / 0%	65.3 / 0%

A single extrapolation number would mislead; what a reader needs is the shape of the degradation. Figure 2 and Table 3 sweep each axis from well inside the box to well outside. Inside, the amortized model is flat: 0.20–1.06 mm and $\geq 99.6\%$ success across the entire range of both axes. Just past the edge it degrades gracefully— $L=90$ mm (12% past) gives 5.1 mm at 60%; $g=40$ (14% past) gives 2.3 mm at 98%—and then cliffs: $L=120$ mm and $g=60$ are 62 and 65 mm at 0%.

The envelope is asymmetric. Shortening extrapolates further than lengthening ($L=30$ mm, 25% below the box: 2.9 mm at 94%; $L=90$ mm, 12% above: 5.1 mm at 60%), and stiffening further than softening ($g=40$: 98%; $g=10$, 33% below: 56%). Both directions that extrapolate well shrink the workspace; both that extrapolate poorly expand it into regions the training arms never reached. The model generalizes over the morphology axis but not to tip positions outside the union of training workspaces.

The Blind column is the mean-arm signature made visible: best at the box centre ($L=50$ – 60 mm, $g=20$ – 25), worse toward every edge, and never above 57% success anywhere.

5.4. Data efficiency

A single-arm specialist's accuracy as a function of its training-set size is shown in Figure 3: 12.0 mm at 5×10^3 samples, 1.86 at 2×10^4 , 0.78 at 5×10^4 , 0.51 at 2×10^5 , and 0.40 mm at 5×10^5 . Reaching sub-millimetre accuracy on one arm costs that arm $\sim 5 \times 10^4$ samples; reaching 0.4 mm costs it 5×10^5 .

The amortized model spent 5×10^5 samples across a continuous 8-D family—in expectation, zero samples at any specific morphology—and reaches 0.39 mm on morphologies it never saw. Per arm served, its data cost is not a number that can be written down: the family is a continuum. What can be said is that one specialist's total budget buys specialist-level accuracy on every arm in the box.

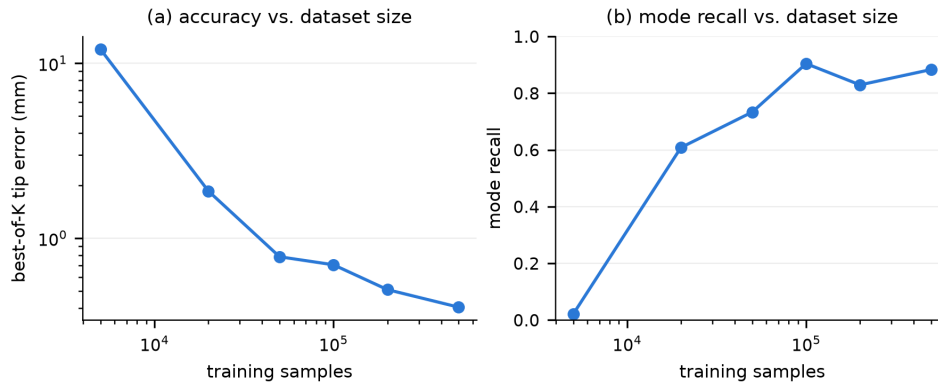


Figure 3. Single-arm dataset-size ablation on the nominal arm (single seed). (a) Best-of-K tip error and (b) mode recall vs. training-set size; both saturate by ~ 0.5 – 1×10^5 samples. The amortized model reaches the 5×10^5 -sample specialist's accuracy on every in-distribution arm with the same total budget spread across the family.

[Amortized model at 10^5 and 2×10^5 total samples, to locate where the amortized curve sits relative to the specialist curve: pending.]

5.5. Multimodality under amortization

Table 4. Mode recall and solution diversity on held-out morphologies, against ground-truth solution sets enumerated on each test arm (2×10^6 sweep, DBSCAN in curvature space; 20 targets per arm, successful samples only). Nominal band is one arm; in-distribution is three held-out arms; extrapolation is five. Seed 0; three-seed $\pm$ pending [3-seed]. Source: `results_morph_modes_seed0.json`.

Band	Amortized		Blind		Nominal	
	recall	div.	recall	div.	recall	div.
Nominal arm	0.90	27.8	0.17	7.8	0.79	28.5
In-distribution	0.94	24.2	0.03	0.0	0.00	0.0
Extrapolation	0.49	6.7	0.02	0.0	0.00	0.0

The single-arm advantage was a multimodality advantage (Table 1). Whether amortization preserves it—whether the model still recovers the distinct solution modes of an arm it never saw—decides whether the amortized model is an IK *solver* or merely an IK *regressor* that happens to be conditioned well. Table 4 answers: it is a solver. On held-out in-distribution arms the amortized model recovers 0.94 of enumerated ground-truth modes, with sampled diversity 24.2 in curvature space—as high a recall as the specialist achieves on the one arm it was trained for (0.79 on this target draw), and diversity in the same range (28.5). The Blind model recovers essentially none (0.03): a model that has learned the mean arm does not merely miss the target, it misses every mode of every arm.

The extrapolation band exposes something a success rate alone hides. At 62% success the amortized model reaches most targets past the box, but its mode recall is 0.49 and diversity 6.7: it finds a solution before it finds *all* of them. Multimodality degrades before accuracy does. A practitioner using the model outside its training family should expect correct actuations but not a complete picture of the alternatives, which matters for the obstacle-avoidance use of Table 1, where the alternatives are the point.

5.6. Only a multimodal model amortizes

Table 5. Architecture comparison under amortization: all three models conditioned on $[\text{tip}, \theta]$ and trained on the same data. Best-of-32 error and success, seed 0. Source: `results_morph_seed0.json`.

Band	Diffusion	MLP	MDN
Nominal arm	0.42 / 100%	13.8 / 1%	18.1 / 2%
In-distribution	0.39 / 100%	14.1 / 3%	16.3 / 5%
Extrapolation	6.2 / 61%	31.8 / 15%	20.8 / 18%

Given the morphology, does any architecture amortize? Table 5 says no. The MLP and MDN, trained on the identical morphology-conditioned data, reach 14.1 and 16.3 mm in-distribution—no better than their single-arm performance in Table 1, and no better than they manage without the morphology at all. The conditioning is not the bottleneck for them; their inability to represent a multimodal conditional is. Amortizing over morphology does not reduce the multimodality of the inverse map—it adds a dimension along which the solution set varies—so the models that could not represent it on one arm cannot represent it on a family. Diffusion’s advantage under amortization is not a capacity advantage: the MLP and MDN are matched in width, and the MLP achieves the *lowest* validation loss of the three (0.264 vs. 0.267 for diffusion). It fits the mean beautifully; the mean is not a solution.

6. The Advantage Is Redundancy, Specifically

If diffusion’s advantage—on one arm or on a family—is redundancy resolution, it should shrink when redundancy is removed and grow when it is added. Both predictions hold on one arm (Figure 4), and the amortization results extend the second.

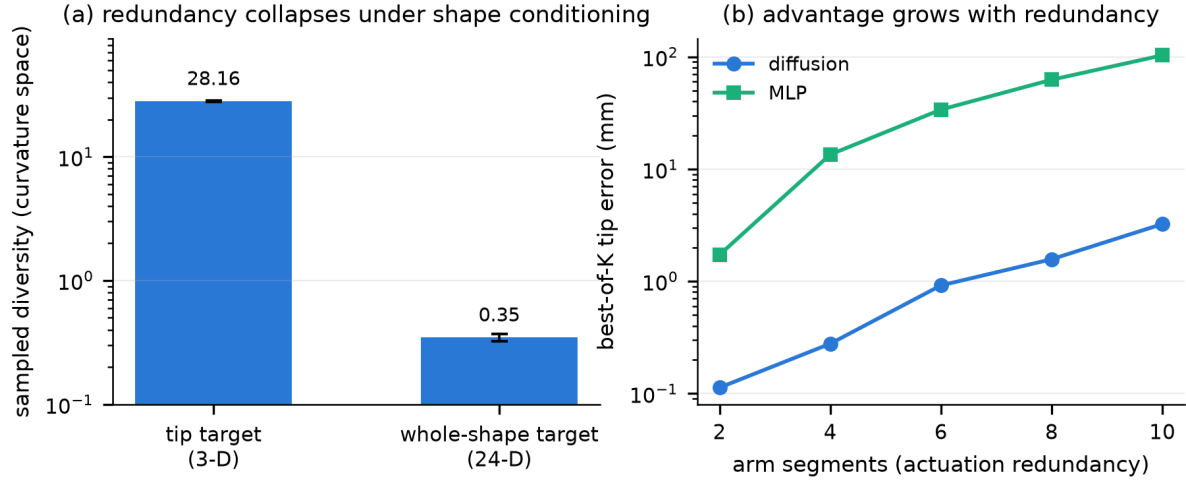


Figure 4. The diffusion advantage tracks actuation redundancy. (a) Replacing the 3-D tip target with a 24-D whole-shape target removes most of the redundancy: sampled diversity collapses $\sim 80\times$ (28.16 ± 0.22 to 0.346 ± 0.022 ; 3 seeds). (b) Increasing the number of segments (single seed per morphology) widens the diffusion–MLP accuracy gap monotonically from two to ten segments; from eight segments on, the MLP’s obstacle-task success is 0% while diffusion’s stays $\geq 99.5\%$.

Removing redundancy. Conditioning on the full 24-D body shape instead of the 3-D tip pins down almost the whole configuration. Diffusion’s sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 ($\sim 80\times$), and the MLP’s accuracy disadvantage shrinks from $\sim 30\times$ to $\sim 3\times$ (0.49 vs. 0.16 mm per-point error, both at 100% success, 3 seeds). When there is one answer, a deterministic regressor is nearly as good.

Adding redundancy. Across two-, six-, eight-, and ten-segment arms (single seed each), the MLP’s error grows monotonically from 1.7 to 103.2 mm with 0% success from eight segments on, while diffusion degrades gracefully (0.11 to 3.2 mm, $\geq 99.8\%$ success). Amortizing over morphology is a third way of adding redundancy—the same tip is now reachable by different shapes on different arms—and Sec. 5.6 shows the same ordering: the deterministic models fail, the multimodal one does not.

7. Under Dynamics-Model Change

Amortization across morphology is generalization across *parameters of the same model*. It is a different question whether a learned inverse map survives a change of *dynamics model*—executing PCC-trained actuations on a Cosserat-rod simulation of the same arm (PyElastica, 48 elements, 1.5 s damped horizon; [18,19]). It does not: accuracy collapses from 0.67 ± 0.03 to 139.9 ± 1.8 mm with 0% success on every seed, and the discrepancy is directional—no couple gain or damping constant aligns the two workspaces.

We characterized what recovers this loss and what does not; Table 6 summarizes and Figures 5 and 6 give the two central curves. Selection under an Elastica query budget improves every target but never solves one; a diversity-aware query ordering significantly beats a confidence ordering (paired $t=3.34$, $p\approx 0.002$); guided sampling is suggestive at best; fine-tuning on 10^4 target-domain samples yields the only nonzero success (67.1 ± 0.9 mm, 10%) at the price of catastrophic forgetting, and mixing source data back in collapses the gain below the first grid point—a retention–transfer cliff with no sweet spot.

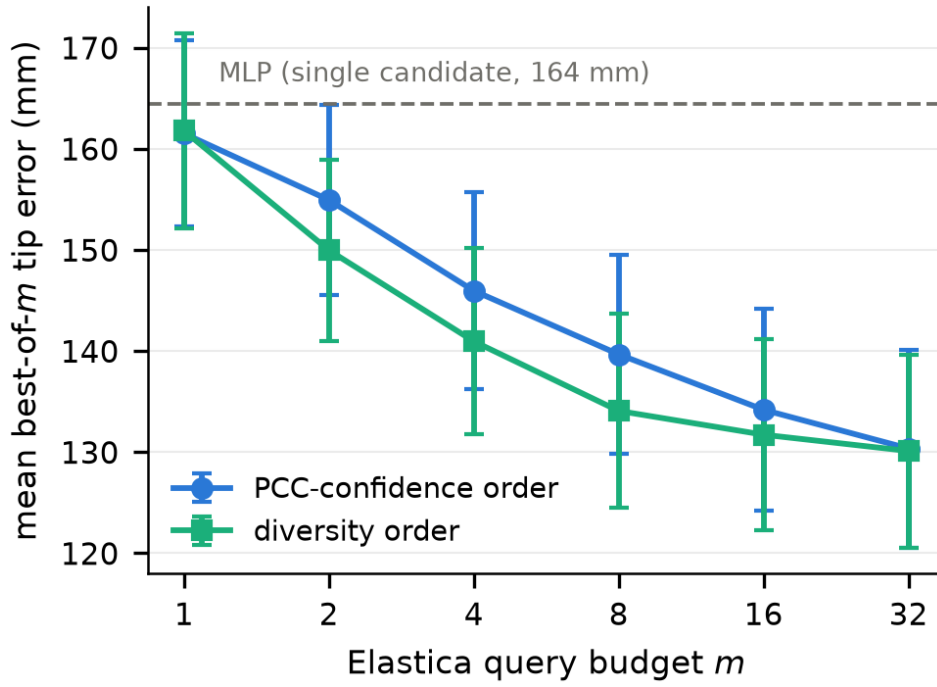


Figure 5. Candidate selection under an Elastica query budget ($n=45$ targets, $K=32$ candidates, mean $\pm$ SEM). A diversity-aware (farthest-point) ordering of which candidates to evaluate beats a confidence ordering; in the paired design the difference is significant at $m=8$ and $m=16$ ($t=3.34$ and 3.21 , $p\approx 0.002$).

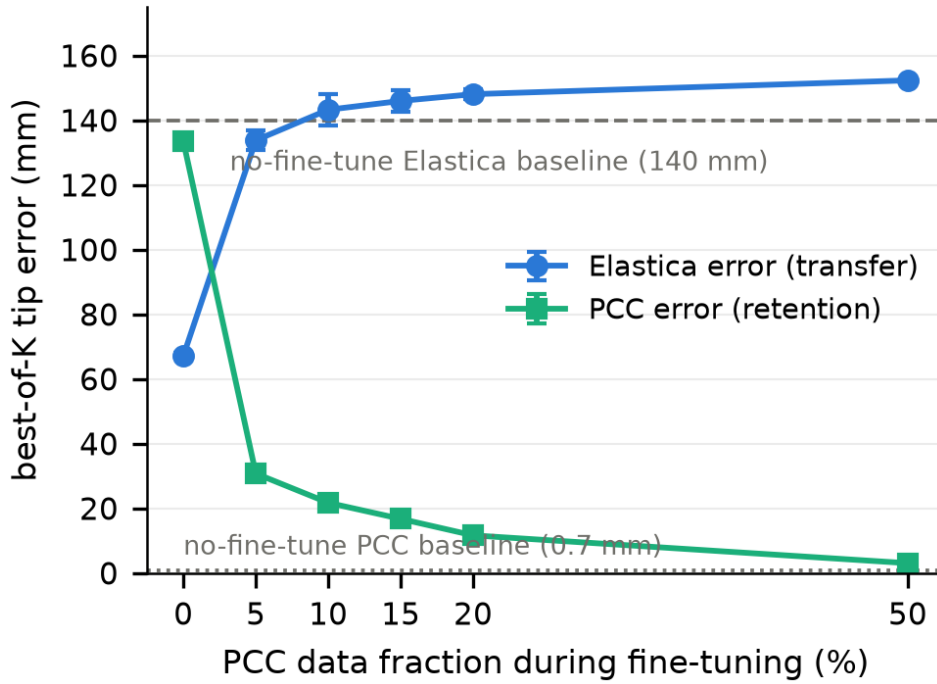


Figure 6. The retention–transfer cliff (mean $\pm$ std, 3 seeds; six mixing ratios). Fine-tuning on 10^4 Elastica samples: retention improves smoothly as source data is mixed back in, but the transfer gain collapses almost entirely by the first grid point (5%). There is no sweet spot.

Table 6. Recovery mechanisms under dynamics-model change (Elastica-domain performance; best configuration of each; each row on its own held-out target set, 15–45 targets).

Mechanism	err (mm)	succ.	Cost / caveat
none (top-1 candidate)	161.5	0%	no target queries
best-of-32 selection	130.4	0%	32 queries/target
+ diversity order	−6.7 at $m=8$	0%	same queries; $p \approx 0.002$
guided sampling	129.0	0%	400 labels; suggestive
gain randomization (blind)	140.6	0%	degrades source; no gain
fine-tune, 0% mix	67.1 ± 0.9	10%	forgets source (133 mm)
fine-tune, $\geq 5\%$ mix	≥ 133.8	$\leq 10\%$	retains source, forfeits gain

The row labelled *gain randomization* is the one this paper reinterprets. It was reported as a null result: randomizing the gain buys nothing. Sec. 5.2 shows why. That model was Blind; it had learned the mean arm, and the mean arm is no closer to the Cosserat arm than the nominal one. Whether an *Amortized* model—one that has learned the inverse map as a function of morphology—can be steered toward the Cosserat arm by inferring an effective morphology for it is the natural next question, and one this paper’s machinery makes askable. It is not answered here.

8. Discussion

What amortization buys. A practitioner with a fabricated soft arm whose segment lengths and gains can be measured—or estimated from a handful of calibration poses—can use one pretrained model rather than train one. The accuracy cost is nil inside the training family and modest for 10–25% beyond it. Because the conditioning is a continuous vector, the model also supports a mode of use no per-arm model can: sweeping θ at a fixed target to ask which morphology in the family reaches it most easily—a design-space query answered in milliseconds.

Why the ablation matters. The Blind result is, we believe, the first controlled demonstration that domain randomization’s failure on this class of problem is specifically an information failure. The practical consequence is direct: when the randomized parameters can be observed at deployment, they should be conditioned on, not marginalized over. DR is the right tool only when the parameters are genuinely unobservable, and its cost in that case—a model that has learned the mean arm—should be budgeted for.

Why only diffusion. The MLP’s lowest validation loss alongside its worst accuracy is the cleanest statement of the paper’s mechanism. A model that fits the conditional mean can fit it arbitrarily well and still return an actuation that is not a solution, because the solutions are a set and the mean of a set is not in it. Amortization makes the set larger.

Relation to rigid-robot amortization. IKDiffuser [4] amortizes over discrete kinematic-tree topologies by tokenization. The soft case differs in that the morphology is a continuous parameter of a single topology, which makes interpolation and extrapolation meaningful notions—and the envelope of Sec. 5.3 is, to our knowledge, the first such measurement for a generative IK model on any robot.

9. Limitations

(1) **Simulation only.** Every result is on a simulated arm; no claim is made about hardware. The motivating fabrication-variability argument is a reason to want amortization, not evidence that this model delivers it on a physical arm. (2) **The morphology must be known.** The amortized model is conditioned on θ ; it does not infer it. On a physical arm the eight parameters would have to be measured or calibrated, and the model’s sensitivity to error in them is not characterized here. (3) **One segment count.** The family varies lengths and gains at fixed $N=4$; segment count changes the actuation dimension and is outside this study’s conditioning scheme. (4) **Statistical coverage.** The single-arm comparison, shape-conditioning, transfer collapse, and fine-tuning grid carry three-seed statistics from the prior study. Of the new results, [Table 2 carries three seeds; the envelope sweep,

architecture comparison, and data-efficiency results are single-seed]. (5) **Extrapolation is bounded.** The model does not generalize to tip positions outside the union of training workspaces; the cliff in Table 3 is a real boundary, not a tail. (6) **Dynamics-model change is not solved.** Sec. 7 is carried over as an honest boundary on what any learned inverse map in this family can currently do.

10. Conclusions

One conditional diffusion model, told which arm it is solving for, solves inverse kinematics for an entire continuous family of soft continuum arms—at the accuracy of a per-arm specialist, with no data from any specific arm, and with graceful degradation for a measured distance beyond the family it was trained on. The same data with the morphology withheld does not: domain randomization’s well-known failure on this problem is an information failure, isolated here for the first time by an ablation that changes nothing but eight conditioning dimensions. Only a model that represents the multimodal solution set can exploit that information, because amortizing over morphology makes the set larger. For learned soft-arm IK, the path past the single-arm model is not more data from that arm but a model conditioned on which arm it is.

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